CHAPTER 5

BUSINESS PROCESS REENGINEERING AND BEST PRACTICES

Quotation

"Business is increasingly digital business." In the future, even more than today, most business will be either digital or depend critically on aspects that are digital. Business thus depends crucially on IT.

Benjamin Grosof

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♣ QUICK LOOK AT CHAPTER THEMES

The previous decade has seen an explosion in projects that pertain to Enterprise Information Systems (EIS). Many types of organization are involved — governments, academia, small- to medium-sized enterprises, large corporations, inter-governmental organizations, and non-profit and non-governmental enterprises. In spite of the massive effort and resources behind these projects, it has been difficult to accomplish these EIS installations on time, within budget, and according to their intended aims. Enterprise information technology provides value to organizations primarily through appropriate business process re-engineering (BPR) — improvement of the current business processes — to ensure that all the business processes and functions are clearly defined, concise and effective.

♣ LEARNING OBJECTIVES

This chapter examines the design of an organization's structure and business processes. It principally focuses on the application of information technologies to renovate organizations. Methods of introducing and implementing information technologies to enable organizational change are examined. It explores the principles and assumptions behind reengineering, examines a case study, and presents alternatives to "classical" reengineering theory. In conclusion, some specific recommendations regarding reengineering are given. In summary, the chapter will:

- ♣ Describe business processes.
- ♣ Present a discussion of how BPR was used in a real case.
- ♣ Pay particular attention to the role of BPR in Enterprise Information Systems.
- ♣ Assess business processes through their key operational characteristics; e.g., efficiency, intended service quality, process flexibility and costs associated with delays, material volume and level of service or product customization.
- ♣ Observe process diagnosis and formulation of improvements including estimating the effects of these improvements in terms of the above process metrics.
- ♣ Model simple business processes in terms of the people and activity sequences involved, the data and materials flowing through those sequences and the dependencies between business information and operational activities.
- ♣ Compare clean slate and technology-enabled business process reengineering.

♣ CASE STUDY

Northwards Housing[1] — Business Process Re-engineering

BACKGROUND

Northwards Housing is an Arms Length Management Organization (ALMO) which was established in December 2005 to manage approximately 12,500 council homes on behalf of Manchester City Council. Following an Audit Commission inspection in autumn 2006, Northwards was judged as a good, two-star organization with excellent prospects for improvement. However, one area that was highlighted as needing significant improvement was that of re-letting of empty properties. In January 2007, Northwards commissioned Enterprise to undertake a root and branch analysis of the voids management process including a detailed review of existing practices, an assessment of how performance was being measured, and the status of partnership arrangements.

INITIAL SITUATION

The focus for Northwards Housing following the Audit Commissions report was the management of their voids properties. The current situation had been highlighted through existing poor performance levels in the related Best Value Performance Indicators. In 2006, the average number of days to re-let a property (i.e. BV212) was between 70 and 75 days, compared with 28 days for top quartile ALMOs whilst rent loss also underperformed at around three percent of total rent roll, compared with 1.21 percent for the top quartile.

KEY CUSTOMER REQUIREMENT

The key deliverable from the project was to identify inefficiencies in the current process and to present a prioritized list of recommendations for future improvement.

OUR RESPONSE

The Enterprise team commenced the Voids Review in March 2006 — the project was scheduled to be completed in an intensive, five-week period and

[1] *Enterprise plc Lancaster House Centurion Way Leyland Lancashire www.enterprise.plc.uk*

in accordance with PRINCE2 methodology. The approach involved collection of qualitative data through a series of workshops, interviews, attendance at meetings, extensive process mapping, and analysis of quantitative data. The review also involved benchmarking activities, including visits to other housing organizations to share best practice and find new solutions to common problems. The project was far-reaching and involved all stakeholders, including tenant representatives, staff at all levels of Northwards Housing, the City Council, and the repairs & maintenance Joint Venture partner, Manchester Working. The Northwards Project Sponsor was kept informed at all times of progress and major findings, and worked closely with the team to ensure the necessary resources were available.

IMPROVEMENTS ACHIEVED

At the end of the Review, Enterprise presented its findings and conclusions to the Northwards Executive Board, together with a series of prioritized recommendations. The final report included both 'As Is' and 'To Be' process maps, detailed recommendations together with impact/resource analysis, an outline Project Plan for implementation of the improvement project, and proposed stepped targets for BV212 together with identification of achievable, cashable efficiency gains. In addition, the Review highlighted a number of relationship and managerial issues that were having an indirect adverse impact on voids performance.

CONCLUSION

Northwards Housing accepted all of the findings and has been fully committed to implementing the improvement project that involves radical change in order to achieve improvement in the voids process. Staff at all levels is now involved in the project that is being championed by both the Chief Executive of Northwards Housing and the Chief Executive of Manchester Working.

♣ SKETCH OF BPR

Reengineering is the deep-seated rethinking and radical redesign of business processes to accomplish dramatic enhancement in significant measures of

performance, such as cost, quality, service and speed.[2] BPR is usually the primary driver in EIS implementation. BPR is frequently used by companies in financial trouble to cut costs and return to productivity. The risk is that the company may jeopardize future growth. "Business processes are methods, steps and activities we perform to do our jobs. For example, in most companies, filling a customer order involves several business processes from processing the order to shipping the product. Some companies have strong business process cultures; others have varying degrees of business process discipline."[3]

♣ INTRODUCTION

Michael Hammer invented the term BPR in 1990 in an article for the *Harvard Business Review* where he challenged managers to obliterate non-value adding work rather than automating it via technology. The article was followed in 1993 with the book *Reengineering the Corporation: A Manifesto for Business Revolution*, written in cooperation with James Champy. The concept of "Business Process Reengineering" traces its origins back to management theories developed as early as the 19th century. The purpose of reengineering is to "make all your processes the best-in-class." Frederick Taylor suggested in the 1880's that managers use process reengineering methods to discover the best processes for performing work, and that these processes be reengineered to optimize productivity. BPR echoes the classical belief that there is one best way to conduct tasks. Accomplishing this goal depends on maximizing the effectiveness of policy development and program delivery, planning and budgetary

[2] Michael Hammer and James Champy are two of the most famous pioneers in the area of business process reengineering, and their 1993 best-selling book, *Reengineering the Corporation: A Manifesto for Business Revolution*, provides the definition of BPR used in this research (p. 32).

[3] Interfacing Technologies Corporation develops award-winning business process management (BPM) software that helps clients map, model and manage business processes and knowledge. Our mission is to empower clients to optimize overall performance and meet compliance regulations by documenting, standardizing, improving and setting clear ownership of their processes, policies and procedures.

arrangements, decision-making processes, organizational structures, workplace relations and people management. Significant gains in performance have been attained through BPR. Increasingly, information and communications technology (EIS) plays a vital role in determining the quality and accessibility of services. Expansion in EIS has opened the door for even greater efficiencies and enhanced service delivery through integrated processes. There are opportunities for even greater gains as public and private sector enterprises transform their existing processes by the strategic application of EIS. For individual enterprises and government agencies, the most noteworthy gains are achieved when EIS decisions are business-driven. Shared arrangements in the private sector have demonstrated considerable benefits — for example, the banking sector's credit card arrangements and the travel industry's booking facilities. Shared arrangements also recommend significant opportunities to improve efficiency and service delivery in the public sector. The strategic use of EIS enables departments and agencies to improve existing program models and introduce new ways of delivering government information and services.

5.1 ALL ABOUT THE BUSINESS AND ITS PROCESSES

Business process reengineering (often referred to by the acronym BPR) is the main way in which organizations become more efficient and modernize. Business process reengineering transforms an organization in ways that directly affect performance.[4] Figure 5.1 shows the evolution of multi-enterprise processes.

5.1.1 Business Processes

A process is a logical set of related activities taking inputs, adding value through doing things, and creating an output.[5] In business, there are many

[4] Peter Carter is Managing Director of Corporate Information Systems Ltd, a UK firm providing consultancy and training services in business process reengineering and associated areas.

[5] Harrington, H. J., Essing, E. K. C. and van Nimwegen, H. (1997). *Business Process Improvement Workbook: Documentation, Analysis, Design, and Management of Business Process Improvement*. New York: McGraw-Hill. p. 1.

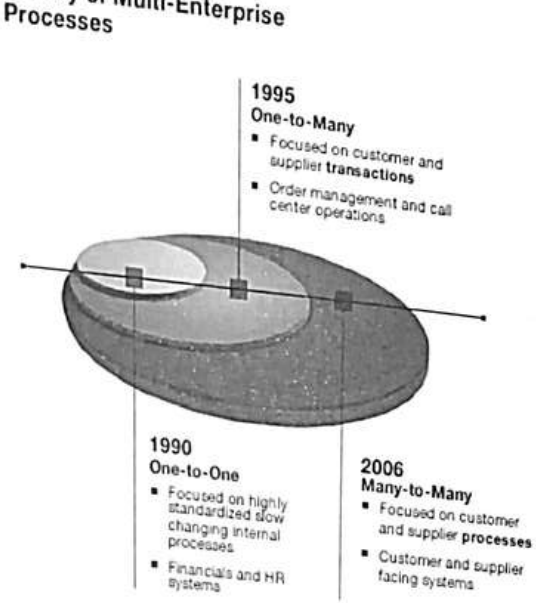


Figure 5.1: Multi-Enterprise Processes.
Source: One Network (2008).[6]

different ways to get work done. Information systems play a key role in providing a means to collect data, store it efficiently, generate reports to let management know what the organization is doing, and archive data for future reference as needed. Figure 5.2 gives a picture of the process of placing an order, showing the many systems involved in such a typical process.

There are, in general, two kinds of processes. Operational processes have to do with accomplishing typical business functions, including product development, order management, and customer support. Infrastructure processes are more administrative, such as establishing and implementing strategy, and managing many aspects of the organization including human

[6] One Network was founded in 2002 with a clear vision: pioneer the use of next-generation technology to enable precision execution of collaborative, multi-enterprise operational processes. Virtual business models have an impact on operations that is matched only by the challenge of managing processes that span so many organizational and IT system silos.

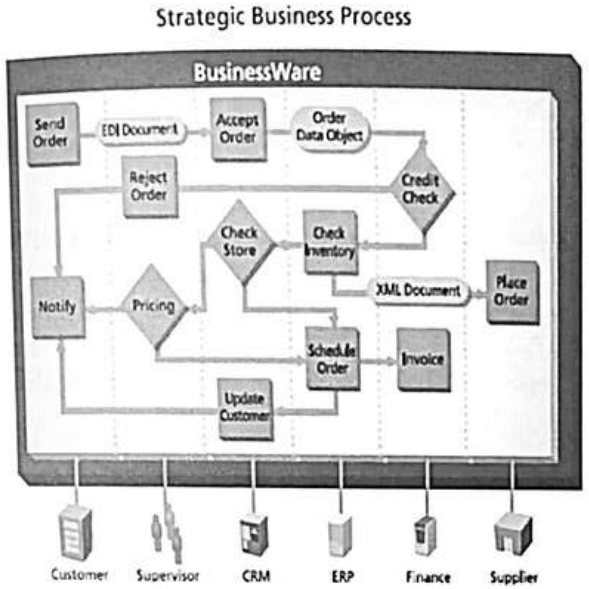


Figure 5.2: Example of a High-Level Business Rrocess.
Source: Vitria (2008).[7]

resources, physical assets, and information systems. Each of these generic processes, whether operational or relating to infrastructure, involves sets of tasks needed to accomplish work. For example, in the operational process of order management, it is necessary to forecast the volume of demand expected for the products produced by an organization. The function of forecasting can be accomplished in many ways:

- Using last month's demand as a prediction for this month.
- Using the monthly demand from a year ago as the prediction for this month.
- Applying a spreadsheet algorithm such as exponential smoothing over available monthly data.

[7] Established in 1994, Vitria combines technology leadership with industry expertise in telecommunications, healthcare and insurance, manufacturing and supply chain, and financial services to dramatically improve strategic business processes across systems, people, and trading partners. www.vitria.com

- Incorporating seasonality indices into such a spreadsheet algorithm.
- Taking known orders and adjusting forecasts based on past demand records.
- Relying upon managerial judgment.
- Using a Ouija board (throwing darts; rolling dice; guessing).

Figure 5.3 shows the typical flow of inputs converted to outputs through a process.

Business process reengineering involves identifying the best way to design the flow and processing of information to obtain forecasts that yield the organization the greatest profit. That requires consideration of

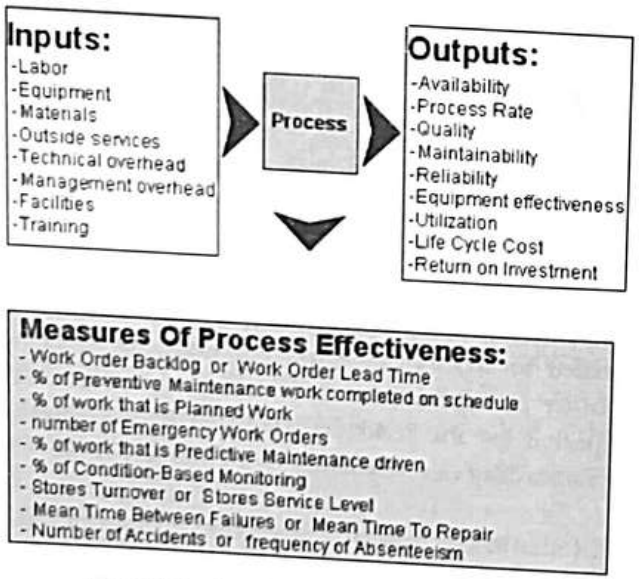


Figure 5.3: Input/Output Diagram of Process.

Source: www.skf.com[8]

[8] SKF @ptitude Exchange is a Web site that provides information to professionals from maintenance technicians to top-level decision-makers, enabling efficient and effective decision-making for improved reliability results. Through SKF @ptitude Exchange, customers gain access not only to SKF's extensive knowledge and experience in rotating equipment, but to a wealth of essential information from SKF's alliance partners as well. These include a range of machinery, component, software, instrumentation manufacturers, and more.

accuracy in avoiding the most damaging kinds of forecast error, and giving the organization the greatest flexibility to respond to risk.

5.1.2 Typical Business Processes

Table 5.1 shows some of the many processes involved in accomplishing operational functions within an organization.

Table 5.2, in turn, lists some infrastructure processes.

Table 5.1: Operational process examples.

Operational	
Function	Process
Develop new products/services	Plan & manage the development process. Research and analyze need. Define customer requirements. Develop & design new product/service concepts. Refine existing product/service concepts. Conduct prototype and market tests. Plan, release, and roll out new products/services. Plan, release, and roll out changes to existing products/services.
Market & sell products/services	Develop & execute market plans. Conduct market & consumer research. Develop competitor intelligence. Manage product/service pricing. Identify and qualify target customers. Develop/maintain customer relationships. Define customer's buying requirements. Develop and propose a solution to the customer. Estimate solution cost and price. Influence customer buying process. Negotiate and close the sale.
Order management	Forecast order volumes. Enter & process orders. Manage customer credit exposure. Plan production/service delivery. Bill and collect revenue.

(*Continued*)

Table 5.1: (*Continued*)

Operational	
Function	Process
Manufacture products and/or provide services	Manage overall production/service requirements & capacity. Plan production/service delivery. Design & engineer customer solution. Produce products and/or provide service. Manage change orders. Plan & manage subcontract services. Control production/service schedule.
Manage logistics	Define logistics strategy. Establish & maintain supplier relationships. Measure & certify supplier performance. Procure materials/services. Manage inventory storage & movement. Manage packing & packaging. Manage outbound logistics.
Provide customer support	Provide customer training. Provide customer interface to the organization. Receive & respond to customer inquiries. Receive & respond to internal inquiries. Dispatch & provide field service/support. Manage return/warranty activities.

Table 5.2: Examples of infrastructure processes.

Infrastructure	
Function	Process
Plan & manage performance	Formulate business strategy & vision. Establish & prioritize goals & objectives. Develop financial/operational performance. Measure & monitor financial/operational performance. Manage business performance. Plan & manage quality & service levels. Design & improve business processes.

(*Continued*)

Table 5.2: (*Continued*)

Infrastructure	
Function	Process
	Design & improve organizational structures & job roles. Manage change. Establish policies & procedures.
Manage finances & accounting	Manage accounts receivable. Manage accounts payable. Manage payroll. Manage the general ledger. Plan & manage taxes. Plan & manage cash. Plan & manage budgets. Plan & manage capital expenditures. Plan & manage risk to corporate assets. Plan & manage loans & equity. Report financial performance. Perform financial analysis & cost accounting. Maintain financial records. Ensure financial control.
Manage human resources	Define human resource needs & skill requirements. Manage the acquisition & termination of human resources. Educate, train & develop employees. Ensure employee communication. Establish reward & recognition systems. Manage compensation & benefits. Provide employee services.
Manage information resources	Define & plan for information resource needs. Develop & enhance data architecture. Develop & enhance communications infrastructure. Develop & enhance software applications. Manage information systems operations. Provide information user support. Archive & dispose of information.
Manage physical assets	Acquire production/service equipment & technology. Maintain production/service equipment & technology. Acquire facilities.

(*Continued*)

Table 5.2: (*Continued*)

Infrastructure	
Function	Process
	Manage & maintain facilities.
	Manage other fixed assets.
Manage support services	Manage public/external relations.
	Perform the legal function.
	Manage administrative services.
	Plan & manage environmental programs.

5.1.3 Reengineering

Reengineering traces its origins back to management theories developed as early as the 19th century. The underlying principle of reengineering is to "formulate all the course of action that is the best-in-class." Frederick Taylor recommended in the 1880's that administrators use procedure reengineering methods to find out the paramount developments for performing work, and that these processes be reengineered to optimize efficiency. BPR stands for the fundamental rethinking[9] and radical[10] redesign of business processes[11] to bring about dramatic improvements in performance.

An illustration will help to explain this.

A typical business process would key in the details into the computer once an order has been received before checking on the customer's credit. Subsequent steps would include taking the goods, from the warehouse, packing and arranging for mode of delivery. None of these, however, is of any value or interest to the customer who is only concerned with the end result of these tasks i.e., delivery of the goods.

What is reengineering? It is not:

- downsizing.
- getting rid of, only unnecessary work jobs.
- merely "restructuring".

[9] Total rethinking begin with the proverbial clean slate and reinvents how you would do your work.

[10] To be radical is to go to the root of things and not about improving what already exists.

[11] Business processes refer to a group of related tasks that together create value for the customer.

- mechanization.
- reengineering an organization or a department in an organization.

The driving force behind reengineering is the 3 C's:

- CUSTOMER
- COMPETITION
- CHANGE

Central to the success of reengineering is the coordination of EIS all the way through the organization. Fundamentally, EIS characterizes the central system of information flow. When organizations improve the central IT processes, such as gathering data only once, integrating cross-functional systems or getting information quickly to customers, radical business process change is achievable. Nevertheless, for IT to be an enabler of reengineering or organizational transformation, it is very important that managers are acquainted with a choice of methods by which EIS can facilitate progress through procedural change. At the same time, administration needs to be conscious of the risks in changing EIS processes.

Reengineering is concerned with questioning the fundamentals of what you do and the services you distribute. Figure 5.4 shows how risk increases with the scope of work involved.

Reengineering works best under the following conditions:

- Case teams should include both managers and those who will do the work.
- Consider reengineering of fragmented processes to reduce obstacles to improving customer service.
- The EIS group should be an essential part of the reengineering team from the beginning.
- BPR should be "owned" throughout the organization, not driven by a group of exterior consultants.
- Top management support is required.
- BPR should be associated with tactical planning, leveraging EIS as a competitive tool.
- A project timetable, preferably between three to six months, reduces the probability of project failure.

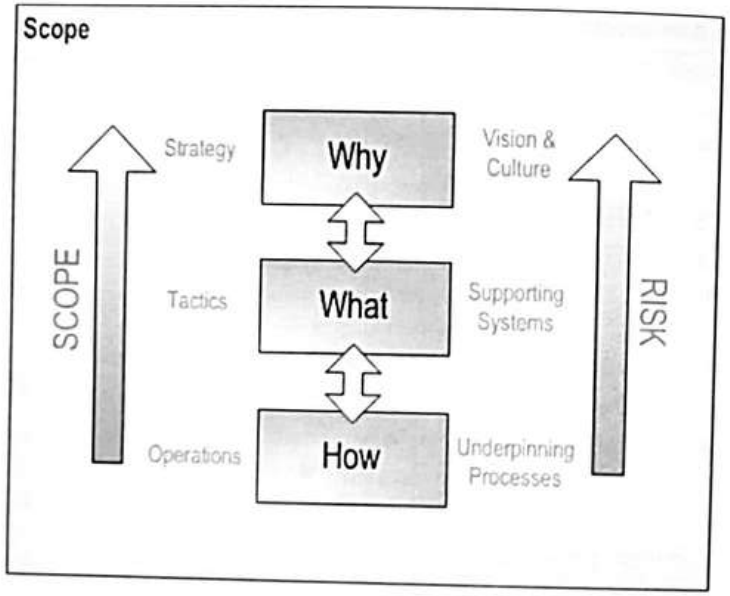


Figure 5.4: Scope of Work and Risk.

Source: JISC (2008).[12]

- Organizational culture should be considered and the BPR process supported by constant communication.

5.1.4 Business Process Reengineering

Contemporary organizations are under great pressure to improve their efficiency while continuing to reduce their costs. The question is not whether to revolutionize, but how to change. BPR is a step in this journey. Hammer[13] and Champy[14] define BPR as "*the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical contemporary measures of performance, such as cost, quality, service, and speed.*" Business process re-engineering (BPR) is being attempted by many firms that are looking for radical gains from the successful redesign of their processes. BPR is a high risk, time-consuming activity, with no guarantee of success, and yet many businesses claim to be re-engineering their processes.[15] Figure 5.5 displays different levels of BPR attainment with respect to process quality design.

[12] JISC infoNet aims to be the UK's leading advisory service for managers in the post-compulsory education sector promoting the effective strategic planning, implementation and management of information and learning technology. The mission of the Joint Information Systems Committee (JISC) is to provide world-class leadership in the innovative use of ICT to support education and research, by funding a national services portfolio and a range of programmes and projects. www.jiscinfonet.ac.uk

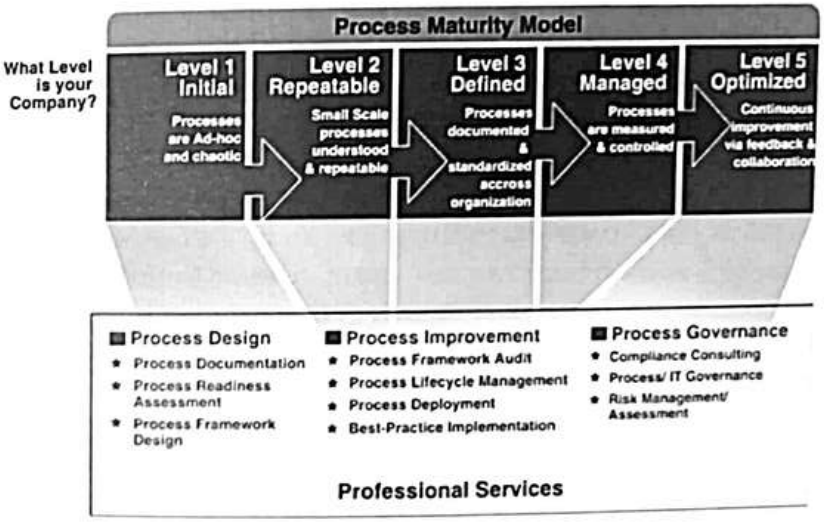


Figure 5.5: Process Quality Design Levels.

Source: APQC (2008).[16]

[13] Dr. Michael Hammer is the leading exponent of the concept of reengineering. He was named by *BusinessWeek* as one of the four preeminent management gurus of the 1990s and by Time as one of America's 25 Most Influential Individuals. He lives in Massachusetts.

[14] James Champy is chairman of Perot Systems consulting practice. He is a leading authority on organizational change and development and business strategy. He lives in Massachusetts.

[15] Named the "Best Online Reference Service" by the CODiE Awards, HighBeam is a premiere online library where you can find research, facts, and articles. It collects millions of articles from newspapers like *The Washington Post* and *The Boston Globe*, magazines like *The Economist* and *Newsweek*, and journals like *JOPERD* and *Journal of Research in Childhood Education* which are all delivered in a single research Web site.

[16] APQC's Process Classification Framework SM (PCF) is a taxonomy of cross-functional business processes intended to allow objective comparison of organizational performance within and among organizations. http://www.apqc.org.

Once an organization decides what form of EIS system they want to adopt, they will next need to specify how the EIS system will be designed. In conventional IS/IT projects, requirements analysis involves a lot of effort to identify what it is that the users of the project need. EIS projects vary highly in the amount of effort needed to identify system requirements. In vendor EIS projects, functionality is given based on vendor research into best practices. (SAP in particular has devoted significant effort to identification of best practices.) In EIS implementations developed with in-house assets, an integrated system serving all of the organization's information system needs must be developed through extensive business process reengineering (BPR).[17] BPR is an effort to identify the best way to do each business task supported by the system. It is an activity in which the way an organization accomplishes each business task is analyzed with the intent of identifying the best way of doing things. BPR is closely tied to EIS, because in order for EIS to provide benefit to the organization, at least some of the ways in which that organization does business must change. Figure 5.6 gives one organization's view of BPR.

Hammer and Stanton noted that reengineering has sometimes been a euphemism for mindless downsizing.[18] They gave an example of how reengineering should work. In the early 1990s, Texas Instruments faced long cycle times and declining sales. BPR was applied, with calculator development accomplished by cross-disciplinary teams from engineering, marketing, and other departments. These teams were to be in control of every aspect of product development from design to marketing. The first pilot teams failed, sabotaged by the existing organization which felt threatened. Functional departments were unwilling to give up good people, space, or responsibility. Power continued to lie in the old functional departments. Texas Instruments responded by changing the way it was organized. Development teams became the primary organizational units. Functional departments focused on redefined missions supporting the product teams. Budget was accomplished by process instead of by

[17] An article on BPR by its most vocal academic proponent is Hammer, M. and Stanton, S. (1999). How process enterprises really work. *Harvard Business Review*, 108–118.
[18] Hammer, M. and Stanton, S. (1999), op. cit.

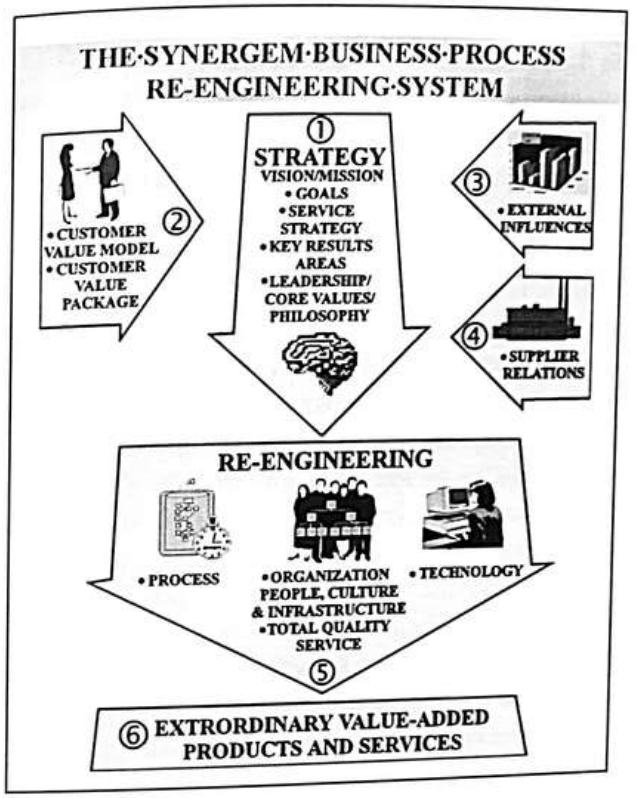


Figure 5.6: An Example BPR System.

Source: Synergem (2008).[19]

department. Office space was reallocated. After the new system became established, new product launching time was cut by as much as one-half, and profitability was enhanced, with return on investment multiplied fourfold.

A business process is what the organization does to get its work done. For instance, almost every organization deals with paying invoices. There

[19] Synergem Management Inc. offers highly experienced & senior management consulting services to private and public sector organizations world wide. Synergem's innovative structure is designed to foster the creativity, responsiveness, quality, breadth, management, accountability, synergy and the administration capabilities required to meet the changes in the modern organizational environment.

are a number of actions in this process. Before computer automation, this amounted to the process given in Figure 5.7.

The manual process begins with receiving documents from agencies within the organization requesting that material be purchased. Rules require that the purchasing department obtain quotes from at least three vendors, and select the low price quote. If the low price quote is above $1,000, managerial approval is required. The agency manager may reject the request and return to the requesting agency, or approve the request in which case the purchase order is processed and the material purchased. If the low price quotation is less than or equal to $1,000, purchasing department staff verify that the purchase request is complete. If not, the request is returned to the requesting agency. If it is complete, the process involves preparation of a paper purchase order, entering this into a master records file, printing the purchase order which is then sent to the vendor, and filing paper documents.

This process involves a lot of expensive managerial time for approval of more expensive invoices. The process itself is well-defined and structured, in that there are clearly stated rules to be followed, which makes automation highly attractive, and that can save the organization a great deal of money. Therefore, a new process might take the from as shown in Figure 5.8.

In the reengineered process, organizational purchasing rules are applied by the computer system. If all required specifications are met, the purchase order is automatically processed, much faster than the old system and with far less people involved, at much lower cost. If all required specifications are not met, the request is returned to the requesting agency in a much faster manner.

This is only one of many ways to perform this business process. Business process reengineering would analyze how a particular operation is performed, and seek better ways to do it, either through more automation, or by adding people to do specific tasks to relieve bottlenecks in the process.

Business process reengineering predates the popular phase of EIS. Most reengineering efforts in the 1980s, which sought more efficient ways to do business, degenerated into wholesale layoffs.[20] The old ways of doing business sometimes remained, but with fewer people to do them.

[20] Hammer, M. (2000). Reengineering redux, *CIO*, 13(10), 143–156.

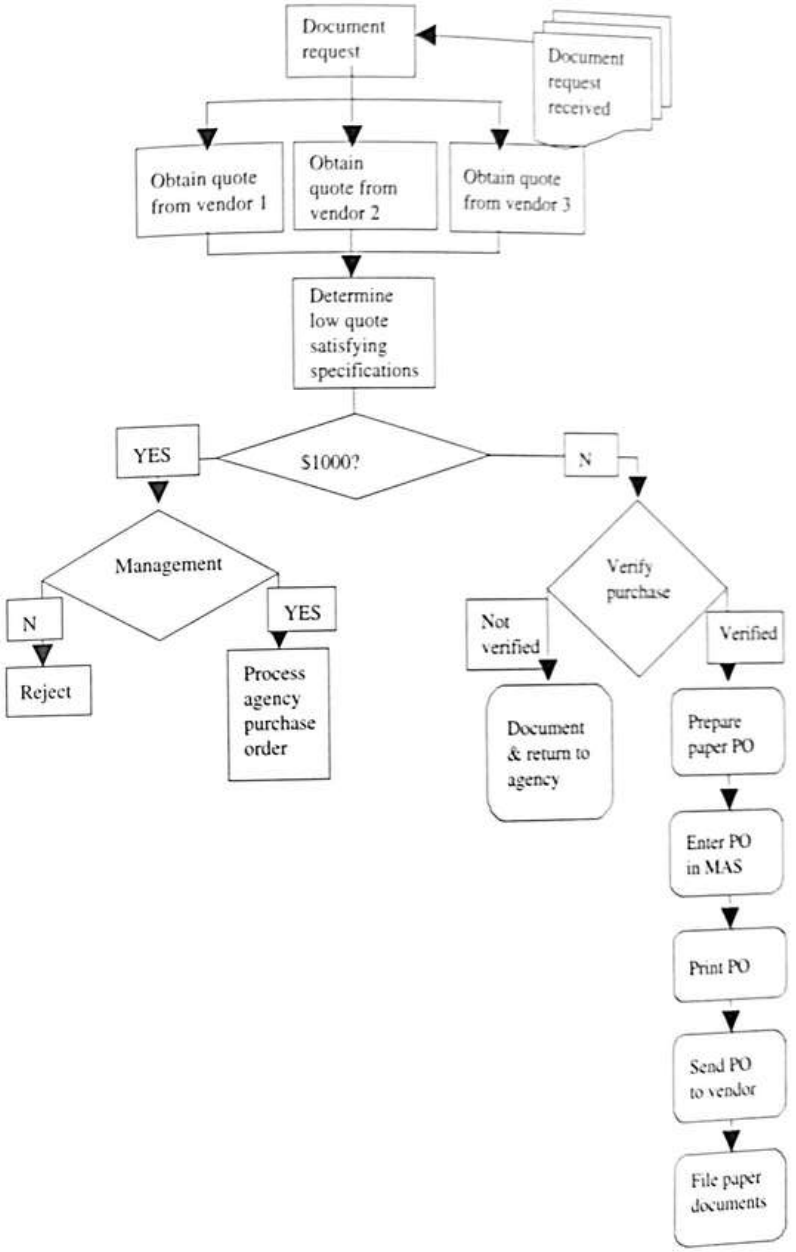


Figure 5.7: Manual Business Process.

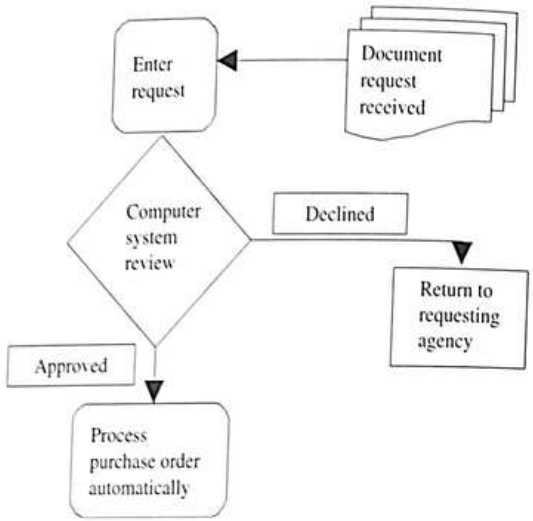


Figure 5.8: Automated Business Process.

Reengineering's primary impact has been credited to be massive reductions in workforce and other short-term cost savings, with less impact on diffusing computer-based automation. EIS software has been credited with rescuing the idea of BPR, forcing companies to redefine and redesign work flows to fit the new software.

Deregulation and competition are drivers for the creation of new business models (BPR) within telecommunications. EIS systems provide a higher level of flexibility in meeting growing customer demands, while demanding higher levels of automation and integration in almost all business processes.[21]

The change implicit in BPR has many risks. Even advocates of BPR cite failure rates of 50 to 70 percent.[22] Sutcliffe reviewed the reasons for the difficulty of implementing BPR:[23]

- Employee resistance to change.
- Inadequate attention to employee concerns.

[21] Levine, S. (1999). The ABCs of ERP. *America's Network*, 103(13), 54–58.
[22] Hall, G., Rosenthal, J. and Wade, J. (1993). How to make reengineering work. *Harvard Business Review*, 71(6), 119–131.
[23] Sutcliffe, N. (1999). Leadership behavior and business process reengineering (BPR) outcomes: An empirical analysis of 30 BPR projects. *Information & Management*, 36(5), 273–286.

- Inadequate and inappropriate staffing.
- Inadequate developer and user tools.
- Mismatch of strategies used and goals.
- Lack of oversight.
- Failure in leadership commitment.

Blanket adoption of an EIS product will discard processes in which the organization has developed a competitive advantage.[24] Instead of changing those processes, the EIS system should be modified. Other activities will be better done following the EIS system's best practices. Even here, a transition period can be expected where employees will have to radically change what they do. Productivity degradation will occur while users learn to adapt to the new system. In the long run the new system is most often better. Those who refuse to adapt to it usually have to learn new skills with their next employer.

The following example demonstrates business process reengineering in an EIS system applied to an academic institution.[25] Babson College implemented a three-year project to transform its business processes. The primary objective was to improve the quality of service delivery to students for admission, records, registration, advising, financial aid, career services, and field-based learning. The college also sought to reduce administrative costs, to redirect these funds to teaching and academic support. Change management teams were formed to reengineer each of the operational areas supported by the system. Critical performance measures were established for each application. The focus was on easy-to-use and effective information systems that students could access themselves. The pre-existing system was a mainframe system with dial-up network and text-based applications. The new system was a multi-tiered client/server infrastructure with 6,000 nodes, over 50 servers, and 1,500 workstations. Data warehousing was used. The system reduced operating costs about 20 percent, half of the planned 40 percent, because some vendor systems were late. One of the problems faced by the information technology team

[24] Schultz, J. D. (2000). Hunt for best practices. *Traffic World*, 41–42.
[25] Kesner, R. M. (1998). Building an Internet commerce capability: A case study. *Information Strategy*, 27–36.

was how to organize access to the system for customers or service providers. Initially it was decided that a graphical electronic mail system front end would be used, with applications written as executables within mail system file folders. But this design was not approved by end users because the mail system was not robust or fast enough for the transaction volumes experienced. The Internet proved to be the solution by providing widespread access. The platform was not as reliable as it needed to be, but it was expected to improve with time.

In an additional example of how best practices can be implemented, Nestle obtained a very large mySAP.com system from SAP for over $200 million in 2000.[26] This system impacted the way in which all 230,000 Nestle employees in 80 countries did their work, because each Nestle employee would have a customized browser-based start page relating to his or her job function to guide the employee to the selected best practice.

Yet another example is Sunoco Products Co., which implemented an e-procurement software allowing the company to reduce purchasing costs for operating resources.[27] The software aided the company in monitoring and enforcing business process changes as part of the EIS implementation. Sunoco was able to reduce its supplier base by 5 percent to 10 percent. They had a goal of a 10 percent reduction in spending on operating resources, which, if attained, would more than pay back their investment.

On the negative side, lack of up-front business process changes has been blamed for installation problems at Farmland Industries Inc.[28] Without BPR, finance and order-entry operations were unable to get the savings they anticipated. This led to the company having to redo BPR and to revise their system implementation.

Within the manufacturing arena, a concept related to BPR is **lean manufacturing**. We will look more at lean manufacturing in Chapter 9. Lean manufacturing is an effort to cut out waste by avoiding activities that don't add value. Throughout a supply chain, manufacturing can often

[26] Konicki, S. (2000). Nestle taps SAP for e-business. *Informationweek*, 792, p. 185.
[27] Shaw, M. (2000). ERP and e-procurement software assist strategic purchasing focus at Sunoco. *Pulp & Paper*, 74(2), 45–51.
[28] Stedman, C. (1999). ERP flops point to users' plans. *Computerworld*, 33(46), 273–286.

involve continuous flows of material without bottlenecks, only producing what the customer has ordered. This reflects a system of demand pull rather than supply push. In the personal computer market, for instance, Dell has a demand-pull system. Some of their competitors had a supply-push system, where product was made in the most efficient way from the perspective of manufacturing. Finished goods would be inventoried in a supply-push system. It appears that Dell's demand-pull approach is outperforming the supply-push approach.

5.1.5 Business Process Management

In today's rapidly changing world, communication service providers are facing a variety of challenges from high capital investments, increasing customer turnover and consolidation of new technologies. These challenges are compelling service providers to focus on processes and their implementation through computerization. New services are being rolled out to customers through new business models and processes to satisfy changing customer requirements, competition and technology advances. This makes the wide-ranging definition of business processes and their effectual management through automation necessary for service providers. The aim of the BPM is to facilitate clients in the areas of business strategy realization, refocusing diverging activities through better coordination, communication and collaboration and finally allocating organizational resources to tackle challenges in a coordinated and efficient manner. Figure 5.9 shows one view of how this might happen.

5.1.6 BPR with Respect to EIS

Proficient business process management (BPM) can facilitate new returns opportunities, trim down operational costs, distribute products and services faster, facilitate customer service, and control operational staff, management, customers, and partners. As more systems and organizations are involved, more process automation gaps materialize, intermingled manual processes are more complex to coordinate, and delays and mistakes become more numerous. Tracking the real-time transactions is very difficult when this information is scattered across different locations.

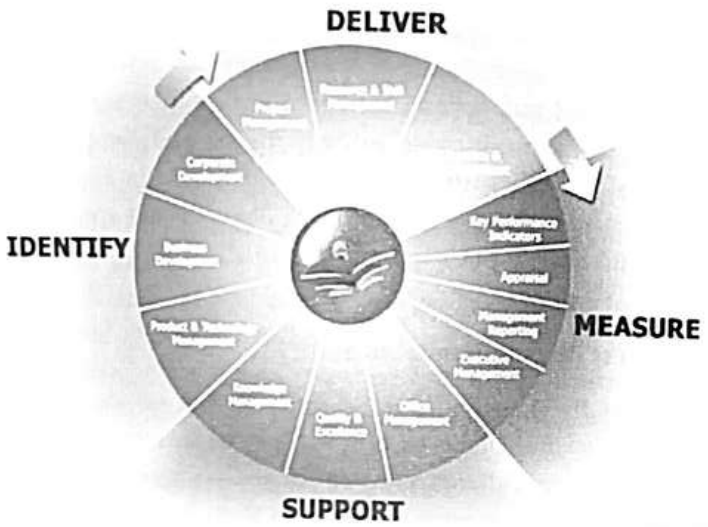


Figure 5.9: Business Process Management.
Source: Pernec (2008).[29]

5.2 BEST PRACTICES IN EIS

One of the primary features of the SAP EIS product has been **best practices**. Business process reengineering is an activity designed to identify a best practice. Once a best practice is identified that would seem applicable to most organizations, it can be incorporated into an EIS system. SAP spends considerable research efforts to identify the best way of doing conventional EIS tasks. They have 800 to 1,000 best practices included in their R/3 software.[30] Consultants often develop further specialized expertise that

[29] Pernec Corporation Berhad (PERNEC) is a Malaysian joint-venture company between Amanah Raya Berhad (Skim Amanah Saham Bumiputra), NEC Corporation (Japan) and Permodalan Nasional Berhad. Its main business activity is to provide Information and Communication Technology (ICT) solutions to customers in Malaysia and abroad. http://www.pernec.com.

[30] Scott, J. E. and Kaindle, L. (2000). Enhancing functionality in an enterprise software package. *Information & Management*, 37(2), 111–122; Dean, C. (2002). ERP Best Practices Checklist. www.deansystem.com.

firms can purchase. A best practice is a method that has been judged to be superior to other methods. This implies the most efficient way to perform a task.

A related concept is **benchmarking**. Benchmarking is comparing an organization's methods with peer groups, with the purpose of identifying the best practices that lead to superior performance. Best practices are usually identified through the benchmarking phase of a business process reengineering activity. Best practices thus often change the organizational climate, and attempt to bring about dramatic improvements in performance.

Vendors attempt to be comprehensive and to be all things to all people. Yet Scott and Kaindle (2000) state that at least 20 percent of the functionality needed by EIS users is missing from vendor packages.[31] There are also many reports of missed deadlines, excessive costs, and employee frustrations in the implementation of EIS. A more participative design approach could help in implementing EIS. If a client implements the entire suite of SAP modules, as well as their tools for system implementation, SAP can ensure timely implementation within budget. However, this approach disregards the human factors of the client business culture.

While business process reengineering was designed to consider human values and business purposes, Taylor states that these factors are clearly neglected in EIS application, and outlines a process emphasizing human factors in EIS implementation. The human factor costs of training and obtaining cooperative participation are key to the successful implementation of EIS.

5.3 REENGINEERING OPTIONS

O'Leary gives two basic ways to implement reengineering: clean slate vs. technology-enabled BPR.[32] While these are not the only choices (they are

[31] Scott and Kaindle (2000), op. cit.

[32] O'Leary, D. E. (2000). *Enterprise Resource Planning Systems: Systems, Life Cycle, Electronic Commerce, and Risk*. Cambridge: Cambridge University Press.

the extremes of a spectrum of reengineering implementation possibilities), they are good concepts to explain the choices available in accomplishing reengineering.

5.3.1 Clean Slate Reengineering

In clean slate reengineering, everything is designed from scratch. In essence, clean slate reengineering involves reengineering followed by selection of the software that best supports the new system design. Processes are reengineered based on identified needs and requirements of the organization. As its name implies, clean slate reengineering has no predefined constraints. This theoretically enables design of the optimal system for the organization. This approach is more expensive than technology-enabled reengineering, but clean slate reengineering is more responsive to organizational needs.

Clean slate reengineering is slower and harder to apply than the technology-enabled approach to implementation. However, clean slate reengineering offers a way to retain competitive advantages that the organization has developed. Ideally, this approach can develop the optimal system for the organization. Clean slate reengineering can also involve significant changes in the way that the organization does business. However, the adjustment in how organization members do their business often retains the features that were found to work well in the past. Thus, while training is required, the impact is probably less than in the technology-enabled approach.

5.3.2 Technology-Enabled Reengineering

In technology-enabled reengineering, first the system is selected and then reengineering is conducted. O'Leary refers to this approach as constrained reengineering. The reengineering process is thus constrained by the selected system. This approach is faster and cheaper than clean slate reengineering, because the software does not have to be changed (it is the basis of the design). Cap Gemini refers to technology-enabled reengineering as concurrent transformation.

The technology-enabled approach designs the organizational system around the abilities of the vendor software. SAP's best practices, for instance, are designed to do things right in the first place. If SAP's research came up with ways to do everything you do better than you used to do them, this would be the best option. It is the easiest to implement, is usually much faster to implement, and thus costs less to implement. On the negative side, it also usually involves the most change in organizational practice, and thus the most complications for training. In practice, therefore, while the EIS installation project looks great from time, budget, and functionality perspectives, the actual benefits to the organization are often disappointing.

O'Leary considered the technology-enabled approach as the most dominant in practice. He cited a survey of SAP R/3 implementers that found that only 16 percent had planned reengineering before they obtained SAP. Of this set of implementers, 33 percent felt that no BPR was needed before implementation, although only 10 percent felt that way after implementation. The most common approach was to undertake BPR simultaneously with implementation of R/3. After the fact, 35 percent thought that reengineering should have come first (the clean slate approach). Table 5.3 compares advantages and disadvantages of both extremes.

O'Leary's advice is that clean slate reengineering should be used by large firms with ample reserve funds. Such firms would have the resources needed and would be more likely to use processes as a basis of strategic advantage. Technology-enabled reengineering should be used by firms that are constrained by budgets or with urgent time requirements. The more standard the processes used by an organization, the more attractive technology-enabled reengineering is. Whichever approach is used, business process reengineering is considered a necessity for firms adopting EIS.

As to practice, O'Leary cited industry surveys of SAP R/3 implementers. The technology-enabled strategy dominated within this group. Firms were surveyed before and after their EIS implementation. Prior to their experience, only 16 percent of those firms surveyed thought BPR should be applied prior to installation of the SAP system. This statistic jumped to 35 percent after the experience. Prior to implementation, 33 percent of

Table 5.3: Comparison of Clean Slate and Technology-Enabled Reengineering.

Clean slate advantages	Technology enabled advantages
Not constrained by tool limitations.	Focus on EIS best practices.
Not limited by completeness of best practice database.	Tools help structure & focus reengineering.
Company may have unique features where vendor best practices aren't appropriate.	Process-bound and thus easier.
Not subject to vendor software changes.	Know that design is feasible.
May be only way to embed processes like Web, bar coding into new technology.	Experience of others ensures design will work.
Maintain competitive advantage.	Greater likelihood of cost, time achievement.
	Software available (already developed).
Clean slate disadvantages	**Technology enabled disadvantages**
No pre-existing structure to design.	Reengineering limited by tool.
Greater likelihood of infeasibility.	System evolution possibly limited by technology.
May involve more consultants.	No relative advantage (others can purchase same system).
May be more costly, slower.	All best practices may not be available.
May not work with selected EIS.	

Source: O'Leary (2000).

those surveyed thought that BPR was unnecessary. After their experience, only 10 percent felt the same. Thus, even if the technology-enabled strategy is adopted, it seems clear that BPR is needed.

Many organizations find it difficult to switch from old legacy systems to EIS. These legacy systems included distribution, financial, and customer service systems developed in-house over the years.

5.4 GENERAL BENEFITS FROM USING EIS

EIS/ERPs have become the driving force for a new business and economic paradigm with far reaching effects for all types of industries

and for the competitiveness of countries. The application of information and communication technology is expected to affect profitability, productivity and employment levels. International comparisons are becoming increasingly significant as the employment of ERP is commonly considered to be a critical factor contributing to national performance on both micro- and macro-economic levels. Benefits can include:

- Enhanced communication through quicker and easier reporting.
- Superior management through electronic data transfer.
- Better networking opportunities, including greater social contact and support.
- Higher quality and accuracy of records.
- Greater access to information.
- Superior efficiency throughout the enterprise.
- Better flexibility for when and where tasks are carried out.
- Facilitates sharing of resources, expertise and advice.
- Less paperwork, with associated reductions in tasks such as filing and photocopying.
- Using templates, pro-forma and shared teaching resources reduces duplication of effort.

The significant benefits while using an ERP Systems can be itemized.

1. Appreciably enhanced service to customers (i.e., individuals, companies and staff) all the way through:
   - The development of customer-centric systems.
   - Amplified provision of self-service capabilities.
   - Redesigned processes leading to enhanced efficiency and competency within the enterprise.
2. Improved policy and decision-making through:
   - Well-timed and precise information.
   - Mechanisms to monitor and deal with the system.
   - Aptitude to allocating information and making decisions on preceding skill.

3. Enhanced deliverance of ERP services and projects via:
   - Formal planning and prioritization of processes in place.
   - Amplified focus of IT resources on core competencies and requirements.
   - Outsourcing non-core activities (e.g., helpdesk, network support, etc.).
   - Reassignment of staff.
4. Long-term gains in competence within IT through:
   - Reduced support requirements for new applications.
   - Standardized desktops, hardware and software.

5.5 PR/BPM SOFTWARE

Business Process Management (BPM) or BPR software is a fast-growing segment of the enterprise software market, due to its support of communal re-engineering. Using BPM software tools, business people can document workflow and processes in their enterprises, to identify bottlenecks and other impediments to effectiveness, and recommend alternative and improved business processes. The purpose of BPM software is to shore up the documentation, analysis, monitoring, and re-design of the business processes in an enterprise. Although tools cannot offer any magic "reengineering button," **business process reengineering and management software/tools** offer essential features, e.g.,

- Documentation of any category of system objects.
- Cross-referencing of information and material related to business processes and tasks.
- Automatic cost aggregation to the superior level.
- Method-driven interface between business process modeling and data modeling.
- Reusage of all identified components for future business processes.

Casewise[33] is a globally recognized leader in Business Process Analysis, Enterprise Architecture, Service-Oriented Architecture and IT Governance & Compliance.

5.6 CONCLUSION

BPR is an important philosophy. It aims to achieve improvements in performance by redesigning the processes which an organization operates, maximizing their value-added content. This approach can be applied at an individual process level or to the whole organization. Business process reengineering is often a major component of an EIS installation. This implies massive changes in the way in which organizations do their business. This has great potential payoff, but also implies a great deal of change in people's work lives, which requires a lot of attention to demonstrating benefits, as well as a great deal of retraining.

Requirements analysis is important in identifying what a proposed system is to do. In EIS projects, requirements analysis takes the form of business process reengineering, to identify the best way (best practice) for each business process supported by the system. There are two extremes in the many ways in which business process reengineering can be accomplished. Clean slate BPR starts from scratch, and is the ideal approach. Technology-enabled BPR begins with the software selected. This is faster and less expensive, as many of the processes are selected from the system. In practice, neither extreme is necessarily best. Hammer and Stanton credited reengineering as doing a great deal of good, despite being a euphemism for mindless downsizing by some.[34] BPR has enabled companies to operate faster and more efficiently, and to use information technology more productively. Employees often obtain more authority

[33] Casewise is a market leader in the provision of software and consultancy solutions to over 3,000 major global organizations for Business Process Modeling, Enterprise Architecture, Service-Oriented Architecture and IT Governance & Compliance solutions.

[34] Hammer and Stanton (1999), op. cit.

and a better understanding of the role their work plays for the organization as a whole. Customers get higher-quality products and more responsive service. Shareholders obtain larger dividends and higher stock value because BPR reduces cost and increases revenues. Executives no longer see their organizations as separate entities, but instead see them as related elements in larger systems linked through information flows across the business, reaching customers and suppliers.

The aim of Business Process engineering within EIS is to

- improve service levels.
- reduce operational cost.
- enhance operational flexibility and efficiency.
- expand customer satisfaction resulting in higher customer retention.
- decrease the cost of customer retention.
- reduce cost of development and maintenance.
- retain revenues.
- reduce 'Time-to-Market,' resulting in faster rollout of new services and features.
- reduce response times, resulting in faster reactions to competitive moves.
- enhance process agility.
- gain improved scalability of processes/systems.
- easier mergers and acquisitions.

To be successful, BPR projects should be top down, cover the entire organization, and include end to end processes. BPR projecs should be supported by tools that make processes easier to follow and study.

We can say that BPR is about fundamentally changing the way firms conduct business. Integrated technology can be a key enabler in this change, but should not be the primary focus of BPR projects.

The following are the vital principles underlying these recommendations:

- **Accessibility:** Ensuring quality information is accessible when required to all staff within the enterprise and external users of the management information.
- **Allocation and collaboration:** Providing information sharing and collaboration and ensuring information is only provided to relevant parties.

- **Amalgamation:** Providing information sourced from multiple systems in an integrated manner across business units and to exterior parties.
- **Competency:** Eliminating information duplication and "re-inventions of the wheel," allowing staff to learn from previous experiences within the enterprise.
- **Ownership:** Development of a culture of information and data ownership to assure improved quality and sharing of information.
- **Relevancy:** Increasing the fit between requirements and information presented by focusing on user requirements and key performance indicators.
- **Unity:** Using common tools and processes for recording, storing and distributing information.

☆ KEYWORDS

- ☆ **Activity:** A process, rationale or assignment that occurs over time and has identifiable results. Activities combine to shape business processes.
- ☆ **AS-IS Model:** A model that corresponds to the contemporary stage of the organization modeled, without any specific improvements included.
- ☆ **Benchmarking:** Comparison of organizational procedures with those of peer organizations.
- ☆ **Best practices:** The set of best ways to accomplish business processes.
- ☆ **Business Process Management (BPM):** An extension of the work flow management movement of the 1990s to include the Business Process Reengineering (BPR) concept of process improvement.
- ☆ **Business Process Reengineering (BPR):** Analysis of the set of tasks making up a business process with the intent of identifying best way of accomplishing it.
- ☆ **Clean slate reengineering:** BPR conducted from scratch.
- ☆ **Conceptual design:** Process of developing a model of what the system should do.
- ☆ **Concurrent transformation:** Synonym for technology-enabled reengineering.

- ☆ **Formal specification:** Output of requirements analysis.
- ☆ **Lean manufacturing:** BPR applied to cut waste in supply chains by eliminating non-value-adding activities.
- ☆ **Process:** Logical set of related activities taking input, adding value through doing things, and creating an output.
- ☆ **Redesign:** The transformation of a business process to achieve significant levels of improvement in one or more performance measures relating to fitness for purpose, quality, cycle times, and cost by using the techniques of streamlining and removing non-value added activities and costs. Redesign projects typically take about six months to complete.
- ☆ **System:** From the Latin and Greek, the term "system" means to mingle, to set up, and to place together. A system is a collection of interrelated elements comprising an amalgamation in totality. A system typically consists of components (or elements), which are connected together in order to facilitate the flow of information (en.wikipedia.org).
- ☆ **Technology-enabled reengineering:** BPR conducted after system is adopted, so that BPR is constrained by system features.
- ☆ **To-be-model:** Models that are the outcome of applying expansion opportunities to the current (AS-IS) business environment.
- ☆ **Validation:** Process to ensure that a valid set of requirements have been developed.

☆ TERMINOLOGY REVIEW

Review Questions

1. Describe business processes.
2. Define business process reengineering.
3. Search the library and/or Internet for business process reengineering practice.
4. Identify risks involved in adopting BPR.

5. Identify a specific business process, and analyze it with the intent of improvement through the use of information technology.
6. Describe the concept of best practices.
7. Search the library and/or Internet for best practices in EIS.
8. Describe benchmarking, and its relationship to EIS.
9. Search the library and/or Internet for benchmarking practice in EIS.
10. Contrast clean slate reengineering with technology-enabled reengineering.
11. Search the library and/or Internet for use of business process reengineering, especially associated with EIS.
12. Understand what and how business processes can be radically improved, dramatically reducing process cycle time and cost, improving the quality of the process products or outcomes.
13. Emphasize challenges and opportunities for BPE/BPR across the process life cycle using Electronic Commerce and Intranet and Extranet technologies.
14. Understand and apply knowledge-based concepts, techniques, and tools for BPE/BPR centered on Electronic Commerce and network information system applications.
15. Was the ROI analysis believable or not? Why? What should be included in the analysis?
16. Can you identify one or more potential business redesign heuristics in the case?
17. What are the basic skills for analyzing and improving business methods, procedures and systems so that they are able to take part in a business re-engineering project in a company for the purpose of productivity improvement?
18. How a working knowledge of the business re-engineering tools and methodologies in the business organizations in terms of business automation, value, processes and risks.
19. What are the abilities needed to select the essential elements and practices needed to develop and implement the business reengineering projects?

20. What are the abilities needed for the analysis and evaluation of the best practices of the business reengineering projects?
21. How do you understand the methods used to changing processes and the impact it has on company culture, people and the integration with other information systems?
22. Develop a strategic plan of IS/IT to support a business strategy aimed at providing an organization with competitive advantage.
23. How do you manage information as a resource and have an understanding of systems development lifecycle?
24. How do you implement, evaluate and analyze strategic business change processes?
25. What can the learning organization strategy bring to knowledge-based organizations?
26. Explain how specific departments and units of functions could be involved in the process of order fulfillment for either a product or service order.
27. Explain with two examples why customer problems could result from lack of integration among functional activities in order processing.
28. Explain how the following technologies must be integrated to provide effective supply chain management and e-business: TPS, databases, LANS and telecom.
29. Explain the steps in planning: an integrated database; a business process re-engineering project; and the technology needed to perform B2B effectively.
30. Explain with two examples each, the legal and ethical issues related to use of IT.
31. How does BPR with the help of EIS facilitate relationships with suppliers and customers. Who are the suppliers of your process? How do products get to customers?
32. Why do BPR efforts rarely produce positive results?
33. When the company demands particular EIS software they have to make compromises on the budget because reworking modules and supplying an EIS Software would be expensive. This is because of the complications involved in doing the same. What is

☆ COMPETENCY REVIEW

1. A functional area is served by an information system. Information systems capture, process, and store data to provide information needed for decision-making.
2. When an ERP system is installed, various configuration decisions are made. These decisions reflect management's view of how transactions should be recorded and later used for decision-making. For example, the system can be configured to limit selling price discounts, thus avoiding unprofitable pricing. Justify your answer.
3. What are the compelling reasons to do this? Is it a "qualifier" in certain industries? Are the efficiencies real? Explain all these point with relevance to ERP and organization.
4. How much are we willing to change the organization? Are these practices really better than what we are currently doing? Do we lose or gain a strategic advantage? If everyone is doing this, how can we gain an advantage? Justify your answers.
5. Provide examples of typical business functions, their related units and activities within those units.
6. Furnish specific examples of how IT has changed: business practices, employee roles, and consumer expectations of business.
7. Give examples of principal and supporting processes. Describe a sub-process and its related procedures and tasks for specified processes.
8. "The BPR exercise throws open an opportunity to give a fresh, radical and global outlook on the entire business process. The organization should try to cash in on this opportunity and come out with a new power packed engine that could propel it into the new millennium." Explain this statement.
9. Why are so many companies still eager to experiment with reengineering, even when they have experienced previous failures themselves?

☆ CHECK YOUR PROGRESS

1. Briefly comment on the following:

   a. BPR has unfortunately been associated with "downsizing" or "rightsizing," both of which mean laying off workers.

   .....

   .....

   b. Business Process Reengineering is approached by examining the strings of processes or cycles.

   .....

   .....

   c. BPR is an excellent tool to use in helping to make work processes faster.

   .....

   .....

   d. The importance of technology as a driver for organizational change.

   .....

   .....

   e. Business Process Reengineering is a significant exercise because it takes the whole business into consideration and relates everything back to delivering value to the customer.

   .....

   .....

   f. An EIS implementation is NOT an IT Project, it is a business rejuvenation project.

   .....

   .....

   g. The BPR exercise throws open an opportunity to give a fresh, radical and global outlook on the entire business process.

   .....

   .....

☆ SUGGESTED READING

Books

Hammer, M. and Champy, J. (2003). *Reengineering the Corporation: A Manifesto for Business Revolution.* Collins Business.

Harmon, P. (2007). *Business Process Change: A Guide for Business Managers and BPM and Six Sigma Professionals*, 2nd Ed. Morgan Kaufmann.

Kalakota, R. and Robinson, M. (2001). *E-Business 2.0.* Addison-Wesley.

Metters, King-Metters, Pullman, Walton (2007). *Business Process Management.* Thomson Southwestern.

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Tsou, D. and Hantos, P. (2005). A case study of a centrally managed ERP implementation for manufacturing plants of a global corporation. *International Journal of Internet & Enterprise Management*, 3(3), 304–311.

White Paper

Nikovski, D. and Kulev, V. (2006). Induction of Compact Decision Trees for Personalized Recommendation. *ACM Symposium on Applied Computing* (SAC).

Online Resources

Digital Opportunities for Development. http://learnlink.aed.org/Publications/Sourcebook/home.htm

Guide to ERPs for Development. www.comminit.com/pdf/CKS_Guide_to_ERPs_for_Development.pdf

http://en.wikipedia.org

Making a Go of It Alone: Some Companies Reengineer without Consultants. Reengineering Resource Center. http://www.reengineering.com/articles/aug96/goalone.htm

NIRMA INSTITUTE OF MANAGEMENT Library AHMEDABAD